Phase 2 - MLflow Experiment Tracking and Model Registry

Project context

This phase extends the Phase 1 fuel-price regression workflow with local MLflow tracking.
The dataset used in all experiments is `global_fuel_prices_2020_2026.csv`, and the prediction target is `petrol_usd_liter`.

Objective

The goal of Phase 2 is to:

? track multiple regression experiments in MLflow
? log parameters, metrics, tags, and artifacts
? compare model performance
? register the best-performing model in the MLflow Model Registry
? compute confidence intervals as a bonus step with `statsmodels`

Dataset and preprocessing

The preprocessing logic is reused from Phase 1 through `pipeline.py`.
The notebook applies the same core steps:

? automatic date feature extraction
? dropping leakage-related target alternatives
? numeric imputation and scaling
? one-hot encoding for categorical variables
? identical train/test split strategy

This guarantees consistency between Phase 1, Phase 2, and the later Phase 3 dashboard.

Tracked models

The Phase 2 notebook compares five configurations:

1. LinearRegression
2. Ridge with alpha = 0.5
3. Ridge with alpha = 5.0
4. Lasso with alpha = 0.001
5. RandomForestRegressor with 300 trees and max depth 12

The notebook evaluates all runs under the same preprocessing pipeline.

MLflow logging

Logged parameters

Examples of logged parameters:

? `test_size`
? `random_state`
? number of numeric features
? number of categorical features
? model-specific hyperparameters such as `alpha`, `n_estimators`, or `max_depth`

Logged metrics

The notebook logs at least the following evaluation metrics for every run:

? `MAE`
? `RMSE`
? `R2`

Logged tags

The following tags are explicitly written:

? `algorithm`
? `dataset_version`
? `target`

Logged artifacts

The notebook logs multiple artifacts, including:

? residual distribution plots
? actual-vs-predicted plots
? `conf_intervals.csv`
? `prediction_ci.csv`
? confidence interval visualizations

Bonus: confidence intervals

As requested in the assignment, the notebook fits a `statsmodels` OLS model on the scaled numeric features and computes 95 percent confidence intervals.

Generated files include:

? `conf_intervals.csv`
? `prediction_ci.csv`

These artifacts are intended for later analysis and for reuse in Phase 3.

Best model and registry

According to `model_meta.json`, the best model selected in Phase 2 is:

? `RandomForest_300trees`

The notebook then registers the best model in the MLflow Model Registry using:

? `BestFuelRegressor_v1`

and transitions the newest version to:

? `Production`

Result summary

Phase 2 is largely implemented and fulfills the coding requirements of the assignment:

? local MLflow experiment setup
? multiple tracked runs
? parameter logging
? metric logging
? artifact logging
? tag logging
? model registration
? production-stage transition
? bonus confidence interval logic

Submission status

The following deliverables are present in code or generated-output form:

? `phase2.ipynb`
? `pipeline.py`
? `mlflow.db`
? `conf_intervals.csv`
? `prediction_ci.csv`
? `model_meta.json`

The following parts are still manual submission tasks:

? `phase2.pdf` screenshots of the MLflow UI
? insertion of explicit screenshots if required by the teacher

Screenshot placeholders

Insert the following screenshots into the final submission if needed:

1. MLflow experiment overview page
2. run comparison table
3. registered model page
4. production model version page

Conclusion

Phase 2 successfully extends the Phase 1 solution into an MLOps workflow with experiment tracking, model comparison, model registration, and uncertainty estimation. The notebook and generated artifacts provide a solid basis for the final Streamlit dashboard in Phase 3.